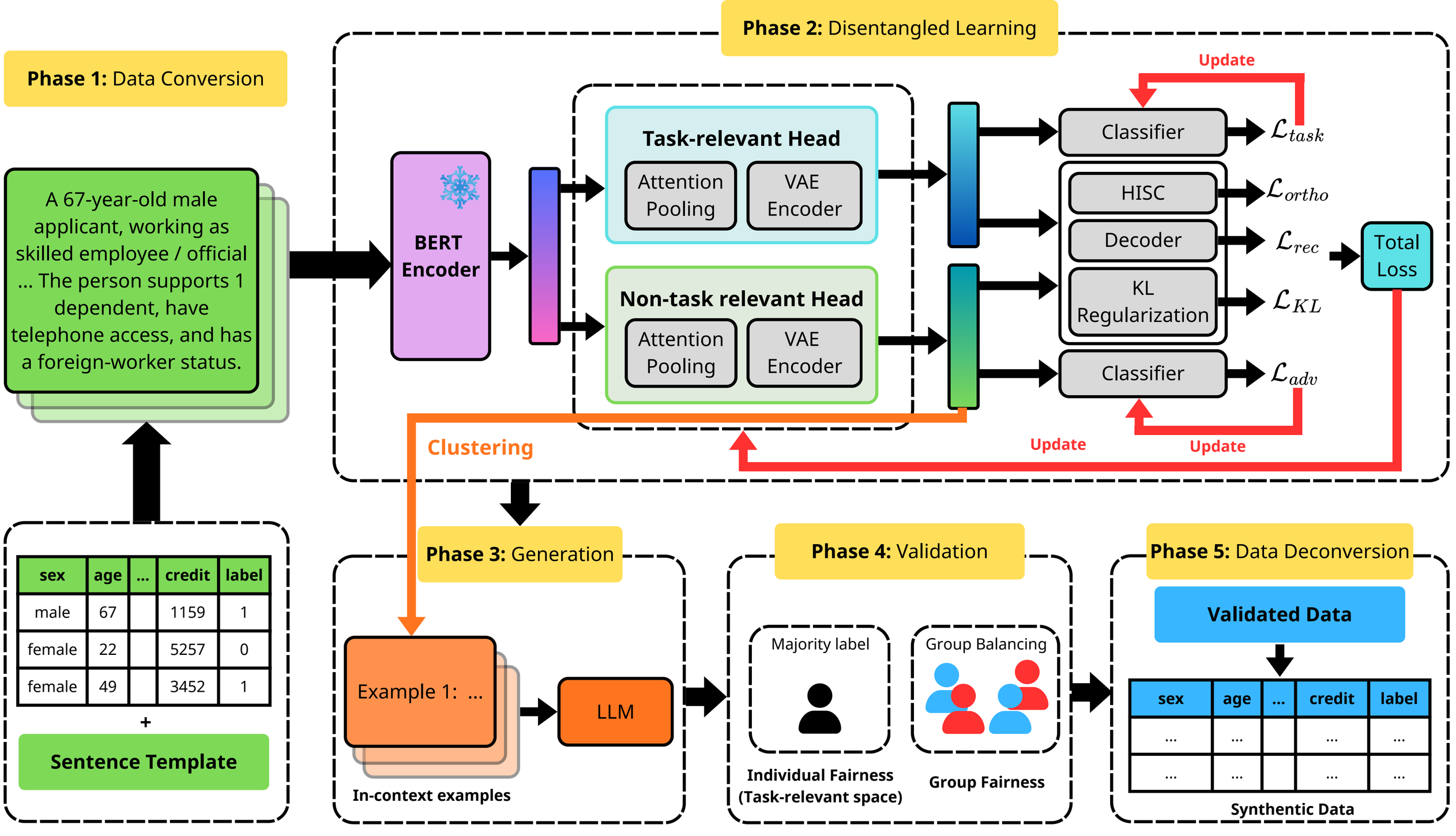

Phase 1: Data Conversion
A 67-year-old male applicant, working as skilled employee / official ... The person supports 1 dependent, have telephone access, and has a foreign-worker status.
sex
age
...
credit
label
male
67
1159
1
female
22
5257
0
female
49
3452
1
+
Sentence Template
Phase 2: Disentangled Learning
BERT Encoder
Task-relevant Head
Attention Pooling
VAE Encoder
Non-task relevant Head
Attention Pooling
VAE Encoder
Classifier
HISC
Decoder
KL Regularization
Classifier
$\mathcal{L}_{task}$
$\mathcal{L}_{ortho}$
$\mathcal{L}_{rec}$
$\mathcal{L}_{KL}$
$\mathcal{L}_{adv}$
Total Loss
Update
Update
Update
Clustering
Phase 3: Generation
Example 1: ...
In-context examples
LLM
Phase 4: Validation
Majority label
Individual Fairness (Task-relevant space)
Group Balancing
Group Fairness
Phase 5: Data Deconversion
Validated Data
sex
age
...
credit
label
...
...
...
...
...
...
...
...
Synthetic Data